

Homework 3

Scott Chiu

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7. Let k be the number of up steps happened before time t , which follows $\text{Binomial}(t, p)$. The probability $P(S_t \geq K)$ can be simplified as

$$\begin{aligned} P(S_0 u^k d^{t-k} \geq K) &= P(S_0 u^k u^{-(t-k)} \geq K) \\ &= P(S_0 u^{2k-t} \geq K) \\ &= P\left(k \geq \frac{\log_u(K/S_0) + t}{2}\right). \end{aligned}$$

Now plug in $S_0 = 4, K = 32, t = 7$ and $u = 2$. The desired probability is

$$P(k \geq 5) = \binom{7}{5}(0.4)^5(0.6)^2 + \binom{7}{6}(0.4)^6(0.6) + \binom{7}{7}(0.4)^7 = 0.096256.$$

8. Let $q = 1 - p$ for simplicity. The tail probability $P(X \geq k)$ can be derived as follows:

$$P(X \geq k) = \sum_{j=k}^{\infty} P(X = j) = \sum_{j=k}^{\infty} q^j p = q^k \sum_{j=0}^{\infty} q^j p = q^k.$$

Now the memoryless property $P(X \geq n + m | X \geq n) = P(X \geq m)$ is true if and only if

$$\frac{P((X \geq n + m) \cap (X \geq n))}{P(X \geq n)} = \frac{P(X \geq n + m)}{P(X \geq n)} = P(X \geq m),$$

which holds given the tail probabilities $P(X \geq m) = q^m, P(X \geq n) = q^n$ and $P(X \geq m + n) = q^{m+n}$.

9. Given a random 10 ml sample, the probability that it is contaminated is

$$\sum_{k=2}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} = 1 - e^{-\lambda}(1 + \lambda) = 0.408167,$$

which we denote by p . The problem is asking for the probability of getting exactly 10 successes in 20 independent Bernoulli trials, each having probability of success p , which is

$$\binom{20}{10} p^{10} (1 - p)^{10} = 0.125017.$$

10. (a)

$$\begin{aligned} \sum_{i=1}^{\infty} P(X \geq i) &= \sum_{i=1}^{\infty} \sum_{j=i}^{\infty} P(X = j) \\ &= \sum_{j=1}^{\infty} \sum_{i=1}^j P(X = j) \\ &= \sum_{j=1}^{\infty} j P(X = j) \\ &= E(X). \end{aligned}$$

(b) Again we denote $q = 1 - p$ and use the tail probability formula for the geometric distribution:

$$P(X \geq k) = \sum_{j=k}^{\infty} q^j p = q^k.$$

By the identity established in (a) we have

$$E(X) = \sum_{i=1}^{\infty} q^i = \frac{q}{1-q} = \frac{1-p}{p}.$$